14. 4 2020

Tutorial XIV

Q1. A coin with $P(H) = p$ and $P(T) = q = 1-p$ is flipped until the first head occurs. Let X denote the number of flips required. Find $H(X)$

SOLUTION

Tutorial XIV

Q1 X represent the number of flips such that

$$p_x = P(X = i) = pq^{i-1}, \quad i = 1, 2, \ldots$$

$$H(X) = -\sum_{i=1}^{n} pq^{i-1} \log pq^{i-1} = -\sum_{i=1}^{\infty} pq^{i-1} \log p - \sum_{i} (i-1) pq^{i-1} \log q$$

$$= -p\log p\,[1+q+q^2+\cdots] - p\log q\,[q+2q^2+3q^3+\cdots]$$

$$= \frac{-p\log p}{(1-q)} - \frac{qp\log q}{(1-q)^2} = \frac{-p\log p - q\log q}{p}$$

$$= \frac{H_2(p, 1-p)}{p} \text{ bits}$$

Note if the coin is fair, $p = \frac{1}{2}$ then

$$H(X) = \frac{1}{1/2} = 2 \text{ bits}$$

Q2. Suppose that one has n coins, among which there may or may not be one counterfeit coin. If there is a counterfeit coin, it may be either be heavier or lighter than the other coins. The coins are to be weighed by a balance. Using information theory, find the minimum number of weighings required to identify the coin.

Q2. Each weighing gives maximum information $= \log_2 3$ bits.
(either the coin is heavier, lighter or equal).

ith coin $\{(i, H), (i, L)\} = 2n$ possibilities
$\quad\quad\quad

Q3. Let $p(x,y)$ be the joint distribution of rvs X and Y

$X \backslash Y$	0	1
0	$\frac{1}{3}$	$\frac{1}{3}$
1	0	$\frac{1}{3}$

Find

(i) $H(X)$, $H(Y)$

(ii) $H(X|Y)$, $H(Y|X)$

(iii) $H(X,Y)$

(iv) $H(Y) - H(Y|X)$

(v) $I(X;Y)$

(vi) Draw a Venn diagram for the quantities in parts (a) through (e)

Q3 The distribution (marginal) of X and Y :

X :	0	1
$p_X(\cdot)$:	$\frac{2}{3}$	$\frac{1}{3}$

Y :	0	1
$p_Y(\cdot)$	$\frac{1}{3}$	$\frac{2}{3}$

Therefore

(i) $H(X) = -\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3} = H(Y) = \log_2 3 - \frac{2}{3} = 1.585 - 0.667$

$= 0.918$ bits

(1)

(b) $H(X|Y) = -\sum_{y=1}^{2} p_Y(y) \sum_{x=1}^{2} p(x|y)\log p(x|y)$

$= +\sum_{y=1}^{2} p_Y(y)\, H(X|Y=y) = \frac{1}{3} H(X|Y=0) + \frac{2}{3} H(X|Y=1)$ (2)

Note

$H(X|Y=0) = -\sum_x p(x|Y=0)\log p(x|Y=0)$

$= -p(x=0|Y=0)\log p(x=0|Y=0) - p(x=1|Y=0)\cdot\log p(x=1|Y=0)$

$= -\frac{1/3}{1/3}\log\frac{1/3}{1/3} + 0\log 0 = 0$

Similarly $H(X|Y=1) = -\left[\frac{1/3}{2/3}\log\frac{1/3}{2/3} + \frac{1/3}{2/3}\log\frac{1/3}{2/3}\right]$

$= -\left[\frac{1}{2}\log\frac{1}{2} + \frac{1}{2}\log\frac{1}{2}\right] = 1$

From (2) $H(X|Y) = \frac{1}{3}\cdot 0 + \frac{2}{3}\times 1 = 0.667$ bits } (3)

Also $H(Y|X) = 0.667$ bits.

(c) $H(X,Y) = -\sum_x\sum_y p(x,y)\log p(x,y) = -\frac{1}{3}\log\frac{1}{3} - \frac{1}{3}\log\frac{1}{3} - 0\log 0 - \frac{1}{3}\log\frac{1}{3}$

$= -3\times\frac{1}{3}\log\frac{1}{3} = \log_2 3 = 1.585$ bit (4)

(d) $H(Y) - H(Y|X) = 0.918 - 0.667 = 0.251$ bits (5)

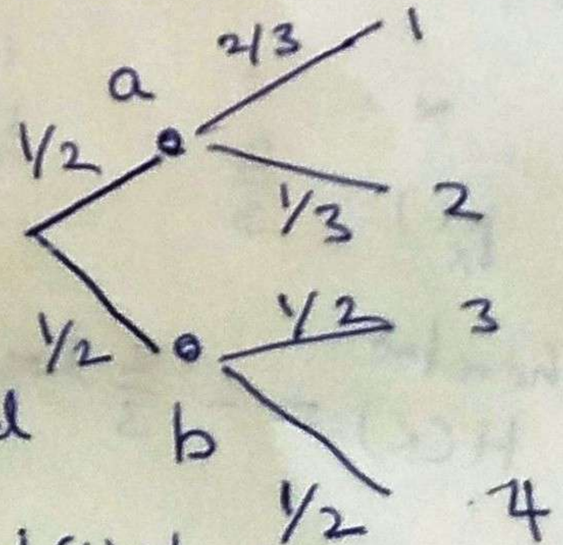
(e) $I(X;Y) = H(Y) - H(Y|X) = 0.251$ bits (6)

Q4. A particle moves along the network as shown. The rv X denotes the position after one second, for which there are two choices, labelled a and b.

and the rv Y is its position after two seconds, for which there are four choices (labelled 1, 2, 3 and 4).

X is a symmetric Bernoulli rv

and we are given the following conditional probabilities for Y



$p_a = (1) = P_{X=a}(Y=1)$, $p_a(2) = \frac{1}{3}$, $p_b(3) = p_b(4) = \frac{1}{2}$

Calculate $H_a(Y)$, $H_b(Y)$, $H_X(Y)$

(e) $I(X;Y) = H(Y)$

Q4 $H_{X=j}(Y) = -\sum p_j(k) \log p_j(k)$, $p_j(k) = P(Y=k \mid X=j)$

– Measures uncertainty about Y when $(X=j)$ has occurred

$H_a(Y) = H(Y|X=a) = -\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3} = 0.918$

$H_b(Y) = -\frac{1}{2}\log\frac{1}{2} - \frac{1}{2}\log\frac{1}{2} = 1.000$ bits

$H(Y|X) = H_X(Y) = \sum_j p_j H(Y|X=j) = \frac{1}{2}H(Y|X=a) + \frac{1}{2}H(Y|X=b)$

$= 0.959$ bits